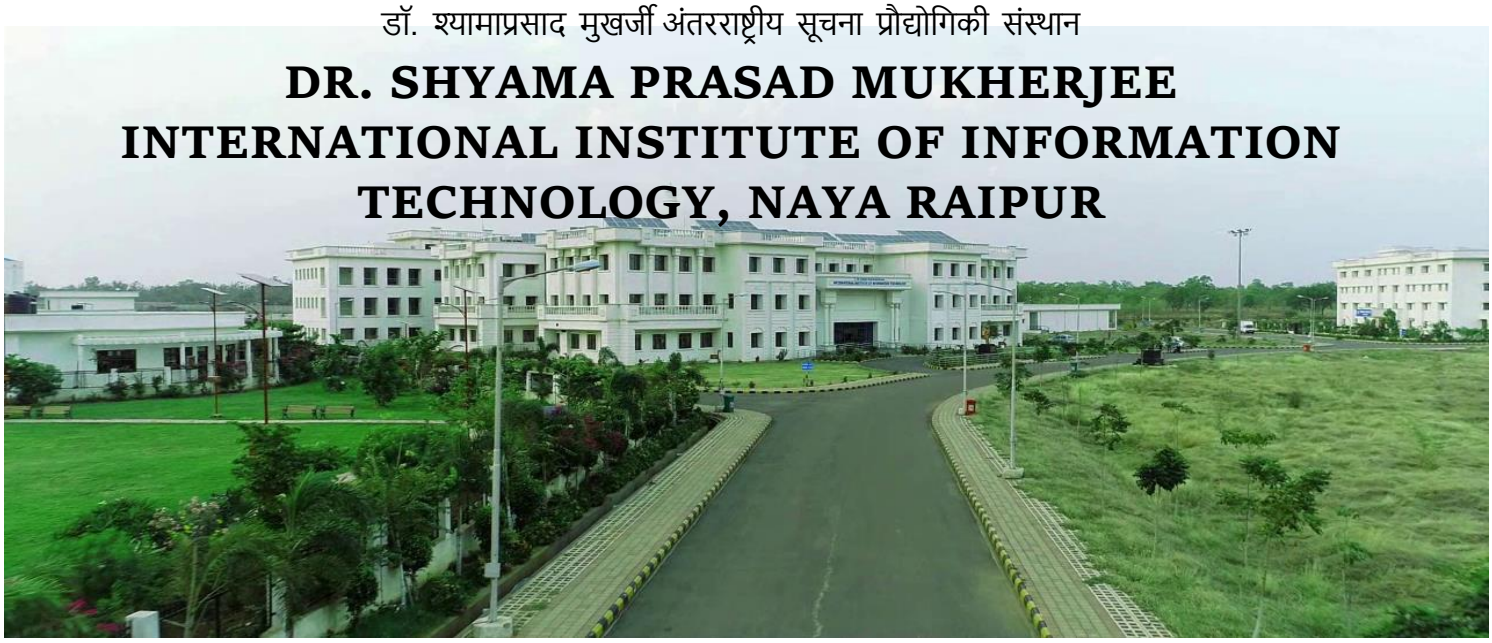
डॉ. श्यामाप्रसाद मुखर्जी अंतरराष्ट्रीय सूचना प्रौद्योगिकी संस्थान

DR. SHYAMA PRASAD MUKHERJEE INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY, NAYA RAIPUR



CENTRES OF EXCELLENCE (COEs)

FOR

STATE-OF-ART EDUCATION AND RESEARCH

AT

IIIT-NAYA RAIPUR

Centre of Excellence (CoE) in Next Generation Networks

Objective:

To promote research and innovation in this highly promising area, IIIT Nava Raipur has setup a 5G testbed in its CoE. The CoE has been established in line with Govt. of India's goal to build indigenous testbeds that closely resemble real-world 5G deployments.

Lab Facilities & Equipment:

This 5G waveform generation and analysis reference solution combine hardware, software and measurement expertise providing the essential components of a flexible 5G waveform generation and analysis test platform. The reference solution enables engineers and researchers to generate and analyze a variety of 3GPP 5G NR waveforms at RF, centimeter-wave, and millimeter-wave frequencies with modulation bandwidths of up to 2 GHz, with an emphasis on 5G 3GPP NR activity in the 28 GHz and 37 - 39 GHz bands. 5G testbed includes Signal Analyzer (2 Hz to 50 GHz), Vector signal generator (250 kHz to 44 GHz), Arbitrary Waveform Generator (14 Bit / 8GSa/s and 12 Bit / 12 GSa/s), Oscilloscope (1 GHz 4 channel) and other basic facilities. IIIT-NR has also developed the anechoic chamber for characterization, configurable software like SystemVue, MATLAB, and Labview. It also has Multi GPU SMP Server to incorporate machine learning applications in communication This will encourage Indian startups and industries to take an early lead in 5G technology. This testbed is first of its kind not only in Chhattisgarh but in Central India. IIIT-NR is now among the few premier institutions in India like IIT-Madras, IISc-Bangalore, IIT-Kanpur, IIT-Delhi, IIT-Hyderabad and SAMEER-Mumbai, which have their own 5G testbed.

Key Achievements:

- Conducted workshop on
    - Robotics & Artificial Intelligence using MATLAB, March 4-6, 2020
    - Evolution of Cognitive Radio Networks and Its Relevance in 5G: From Theory to Practice, Jan 23-26, 2020
    - Workshop on Cognitive radio, Aug 5-7, 2019
    - Physical Layer Techniques of 5G Wireless Communication, May 7-11, 2018
- Participate in telecommunications standards development society (TSDSI), india
- Submitted project proposal worth more than 10 crore
    - NM-ICPS with IIIT Bangalore, IIT Hyderabad, IIT Bhilai, Advanced Communication System
    - Meity with IIT Patna, Design & Development of Energy-Aware 3D-MIMO with Massive Antennas for 5G Enhanced Mobile Broadband using Machine Learning algorithm
    - SERB, Joint parameter estimation and symbol detection in OFDM based IoT Networks: Machine Learning based approaches
- Submitted proposal to conduct workshop in high reputed international conference
    - Orchestration of 5G Network & beyond using Machine Learning (O5ML 2020) in conjunction with WCNC, April 6-9, 2020
- Collaborate with external Industry and academia in India & abroad
    - IIT, IIITs
    - University College Dublin

Artificial Intelligence and Deep Leaning (AIDL) lab

Objective:

AIDL lab established to carries out work on a variety of topics within Artificial Intelligence and Machine Learning with a focus on using real world problems to discover computationally hard problems of modelling uncertainty, learning predictive models and enabling decision making. It also provides computing platform for novel languages like Python, R, and OpenCV and other necessary for building coherent set of ideas, fundamental of Machine Learning, Artificial Intelligence and Deep Learning models and algorithms.

Lab Facilities & Equipment:

It has a Multi GPU Server equipped with processors are high-end 2 x Intel® Scalable processors Xeon 6148 and dual GeForce GTX 1080TI (11GB) with total SSD Storage of server 30 x 4000GB. Lab includes 10 Multi GPU workstations equipped with 2 x Intel® Scalable processors Xeon 4114 with 64GB (Max 1024GB) of memory, 8TB of disk space and dual GeForce GTX 1080TI (11GB).

Key Achievements:

- Conducted workshop on data science, machine learning and artificial intelligence etc
    - ATAL workshop on deep learning
    - Workshop on machine learning and AI
- Singed MOU
    - with industry like Farmguide, Delhi
    - with Indira Gandhi Krishi Vishwavidyalay (IGKV), Raipur
- Ongoing
    - DST, Artificial Intelligence and IoT Based UAV System for Crop Health Monitoring and Forecasting, 24 lakhs, 2020-2022
    - TEQIP, Real-Time Detection, Classification and Remedy of Invasive Insects in Soybean Crop Through Deep Learning Technique
    - TEQIP, Use of Artificial Intelligence and Deep Learning in Identification of Major Disorders Important Crops
    - TEQIP, Use of Block Chain to Assure Quality of Farm Inputs (Seed, Planting Material, Fertilizer Etc)
- Submitted project proposal worth more than 1.5 crore
    - Joint project submission with COEP Pune
    - SERB, 5G, IoT and Artificial Intelligence based wearable devices for health monitoring and forecasting
    - IMPRESS, ICSSR, Target insects early identification and labeling in crop using deep learning techniques: Smart Agriculture
    - IMPRESS, ICSSR, Immediate healthcare support using intelligent accident detection through surveillance video

Internet of Things (IoT) and Sensor lab

Objective:

The IoTs lab provides necessary software, equipment to develop, and characterization of novel smart sensors for IoT applications.

Lab Facilities & Equipment:

Along with these the lab is equipped with various kits like Smart Healthcare, Smart Agriculture, Smart Village and City. The main research is focused on development of a wide range of wearable sensors for pervasive monitoring, self-powered sensor nodes through energy harvesting and implementation of IoT based solution in in the area of healthcare, agriculture, and environment through the introduction of novel sensing platforms and ICT approaches. Through fusing multiple sensing modalities, we aim to develop a smart environment, which can learn and adapt to different environments and provide proactive sensing for different applications.

Key Achievements:

- Conducting workshop on IoT, Sensor etc
    - Robotics & Artificial Intelligence using MATLAB, March 4-6, 2020
    - Sensor network, IoT and Internet of everything, Dec 13-15, 2019
    - FDP on "Internet of Things", July 2018
- Conduct event SCInTFIC annually for school students for past three years
- Ongoing project worth is 2.1 core from Meity
    - "TribeConnect: Integrated Smart Tribal Eco Platform - A Proof of Concept in Chhattisgarh," MeitY CC&BT project , India; Amount: Rs. 210.00 Lakhs; Duration: 2020-2023
    - "Design and development of low cost irrigation system for smart farming," ICAR-RCER
- Submitted project
    - NM-ICPS with IIIT Bangalore, IIT Hyderabad, IIT Bhilai, Advanced Communication System
    - DST, Ageing Fit: An IoT based integrated care for geriatricians in old age homes
- Singed MOU with AIIMS Raipur
- Offered online summer internship 2020.
- Collaborate with external Industry and academia in India & abroad
    - Dr. Saraju P. Mohanty, Smart Electronic Systems Laboratory, University of North Texas, USA
    - Dr. Sudip Misra, SWAN Lab, IIT Kharagpur
    - Dr. Ashok K. Talukder, SRIT, Bangalore
    - Dr. Preetam N. Wasnik, AIIMS Raipur
    - Dr. Akram Ahmed, ICAR-RCER, Patna
    - Dr. G. L. Sharma, IGKV Raipur

Big Data Computing Lab

Objective:

The purpose of the Big Data Computing Lab is to educate students in all aspects of large and distributed information systems (e.g. system development, testing, maintenance, data integration, networking, and application development) and, thus prepare them for highly emerging technologies in fast-growing IT industries such as Big Data computing, Health-care informatics, Data integration and Data analytics.

Lab Facilities & Equipment:

The research in the lab mainly focuses on next-generation data management and mining issues on BigData with special attention to heterogeneous information networks and social computing. An experimental setup for BigData analytics platform primarily using open source technologies such as Hadoop Distributed File System (HDFS), MapReduce, YARN, Hive, MongoDB, Hbase, Apache Spark, MLlib, etc.

Key Achievements:

- Conducted workshop
    - Cyber security and blockchain technology, Nov 28-30, 2019
- Ongoing project
    - Meity, Smart Agriculture for Tribes using Artificial Intelligence and Big Dataworth is 70 lakhs, 2020-2023
    - SERB, Identification of Diseases in Millet Crop using Deep Learning for Next Generation Climate Smart Agriculture, 26 lakhs, 2019-2022

Antenna and Microwave Circuit Design Lab

Objective:

The Microwave Lab supports intermediate and advanced courses in Electromagnetics and Microwave Engineering. Students experiment with transmission line propagation, antennas and microwave circuit components.

Lab Facilities & Equipment:

The microwave laboratories provide the necessary support for training the students in the area of RF and Microwave Engineering. It offers analysis and simulation of various components and devices to understand the basics of RF and microwave engineering, to boost the quality of engineering education, deepen understanding, and provide the necessary practical skills to young minds. The laboratory uses the following software and devices: Computer Simulation Technique (CST) Studio, Ansys HFSS, MATLAB, Vector Network Analyzer, Microwave signal generators, Antenna gain measurements, Waveguide and coaxial components of a wide variant and a anechoic chamber.

Key Achievements:

- Conducting workshop
    - ADS training session tool conducted in Antenna and Microwave Circuit Design Lab on Feb 5. 2020.
    - On Design of Microwave Antennas for Communication System" on Aug 11-12, 2017 using the platform of CST Microwave Studio.
    - On Design of Microwave Antennas/ Filters/ Couplers/ RADAR", March 24-25, 2017 on ADS, antenna, microwave communication etc
- Project Submitted
    - SERB, Design and development of SIW based microwave components for wireless applications

VLSI & Embedded Lab

Objective:

The lab is well equipped with most of the up-to-date industry standard VLSI tools, such as Cadence, Synopsys etc.

Lab Facilities & Equipment:

The hardware resources includes Raspberry PI, Universal FPGA/CPLD Card, XILINX FPGA ZYNQ, NEXYS board, MSP 432 MCU Board, Simple link wireless MCU Board, INTEL Galileo Board Gen 2, ZYNQ-7000 AP SOCZC702 Evaluation Kit, Genesysz (410-300), NEXYS 4 DDR, Kintex 7FPGA KC 705 (EK-K7- KC 705-G) and Zed Board (410-248) etc. The primary objective of the LAB is to provide hands-on experience to students in the field of analog, digital and mixed-signal circuit design where they are able to conduct world class research, produce competitive results and create solutions tailored to society's needs, and to disseminate the knowledge attained for public use while adhering to universal moral values.

Key Achievements:

- Conducted workshop on VLSI and nano-circuit
    - A 5-days workshop was conducted on VLSI and Nanocircuits
- Offer facilities to other institute students in summer

3D Printing & Fabrication Lab

Objective:

The 3D Printing & Fabrication Lab is a state-of-the-art digital manufacturing space and allows students the opportunity to quickly fabricate parts from computer-generated designs to rapidly iterate prototype designs. Digital Fabrication is revolutionary technology that will change the way we fabricate everything, from machine parts, concrete structures, auto-parts, prosthetics, body organs, foods, printed circuit boards, to almost everything.

Lab Facilities & Equipment:

The lab offers access to laser cutting and engraving, CNC cutting, milling, 3D scanners and an array of 3D printers equip to tackle different designs. The lab is always buzzing with activity and new ideas proving vital for students working on projects for both class and research. It is a small scale workshop offering digital fabrication which empowers the users to create smart devices for themselves which can be tailored to local or personal needs.

Key Achievements:

- Conducted workshop on
    - Hands-on/demonstration of 3D printing for IIITNR students in collaboration with TechB Pvt. Ltd.
    - 3-Day Hands-on workshop on Robotics & AI using Matlab", March 2020 (Jointly)
- Designed Automatic Hand Sanitizer using lab facilities in context to COVID-19.
- Submitted two project proposals citing the lab facilities